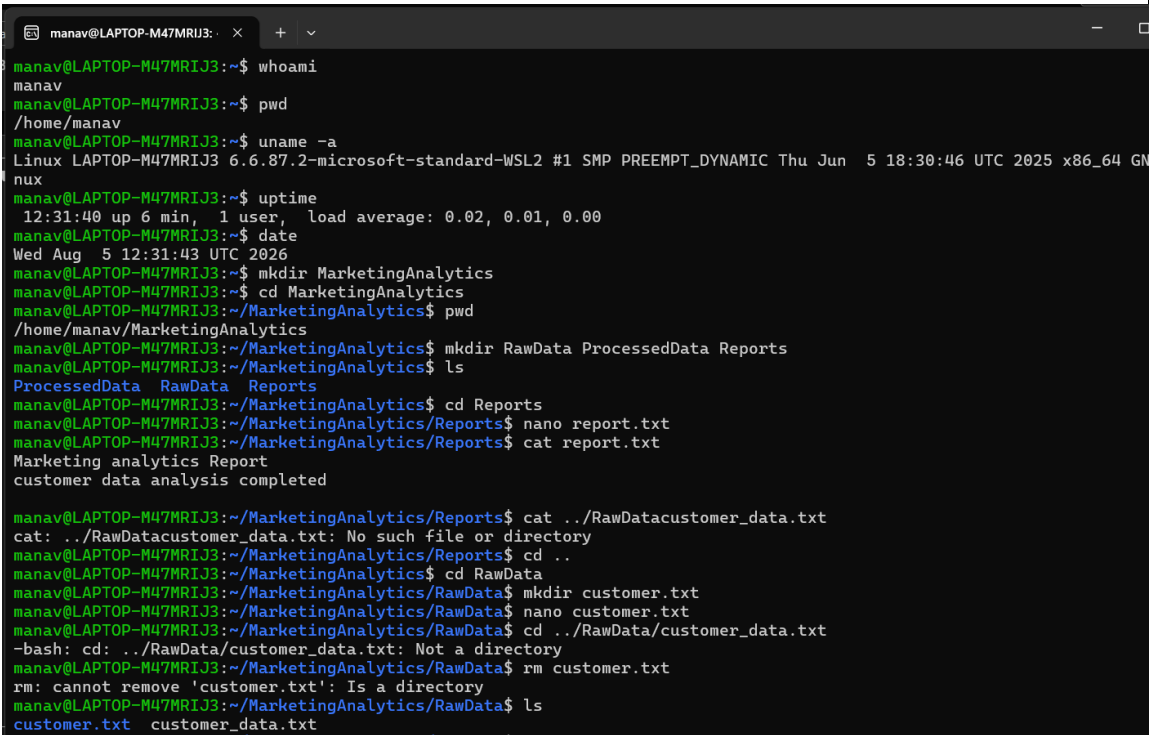


Practical 1	AIM: Linux System Administration and File Management for Marketing Analytics Team
Problem Definition	Scenario: You have recently joined DataAnalytics Inc. as a Linux System Administrator. The company runs its analytics platform on Linux servers where project data, reports, scripts, and user accounts must be managed efficiently. Your manager has assigned you the responsibility of setting up the project environment, managing files and directories, monitoring system resources, securing data, and creating backups for the Marketing Analytics team. Tasks: 1. Identify the current user. 2. Find the current working directory. 3. Display system information. 4. Check system uptime and date. 5. Create a directory named MarketingAnalytics. 6. Create subdirectories: RawData, ProcessedData, Reports. 7. Create data and report files. 8. Add sample content. 9. Display file contents. 10. List all files and directories. 11. Search for report files. 12. Search specific content inside files. 13. Copy report files to Backup directory. 14. Rename files. 15. Move files between directories. 16. Delete a file. Delete an empty directory.
Output	 <pre> manav@LAPTOP-M47MRIJ3:~\$ whoami manav manav@LAPTOP-M47MRIJ3:~\$ pwd /home/manav manav@LAPTOP-M47MRIJ3:~\$ uname -a Linux LAPTOP-M47MRIJ3 6.6.87.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 5 18:30:46 UTC 2025 x86_64 GNU Linux manav@LAPTOP-M47MRIJ3:~\$ uptime 12:31:40 up 6 min, 1 user, load average: 0.02, 0.01, 0.00 manav@LAPTOP-M47MRIJ3:~\$ date Wed Aug 5 12:31:43 UTC 2026 manav@LAPTOP-M47MRIJ3:~\$ mkdir MarketingAnalytics manav@LAPTOP-M47MRIJ3:~\$ cd MarketingAnalytics manav@LAPTOP-M47MRIJ3:~/MarketingAnalytics\$ pwd /home/manav/MarketingAnalytics manav@LAPTOP-M47MRIJ3:~/MarketingAnalytics\$ mkdir RawData ProcessedData Reports manav@LAPTOP-M47MRIJ3:~/MarketingAnalytics\$ ls ProcessedData RawData Reports manav@LAPTOP-M47MRIJ3:~/MarketingAnalytics\$ cd Reports manav@LAPTOP-M47MRIJ3:~/MarketingAnalytics/Reports\$ nano report.txt manav@LAPTOP-M47MRIJ3:~/MarketingAnalytics/Reports\$ cat report.txt Marketing analytics Report customer data analysis completed manav@LAPTOP-M47MRIJ3:~/MarketingAnalytics/Reports\$ cat ../RawDatacustomer_data.txt cat: ../RawDatacustomer_data.txt: No such file or directory manav@LAPTOP-M47MRIJ3:~/MarketingAnalytics/Reports\$ cd .. manav@LAPTOP-M47MRIJ3:~/MarketingAnalytics\$ cd RawData manav@LAPTOP-M47MRIJ3:~/MarketingAnalytics/RawData\$ mkdir customer.txt manav@LAPTOP-M47MRIJ3:~/MarketingAnalytics/RawData\$ nano customer.txt manav@LAPTOP-M47MRIJ3:~/MarketingAnalytics/RawData\$ cd ../RawData/customer_data.txt -bash: cd: ../RawData/customer_data.txt: Not a directory manav@LAPTOP-M47MRIJ3:~/MarketingAnalytics/RawData\$ rm customer.txt rm: cannot remove 'customer.txt': Is a directory manav@LAPTOP-M47MRIJ3:~/MarketingAnalytics/RawData\$ ls customer.txt customer_data.txt </pre>

```

manav@LAPTOP-M47MRIJ3:~/MarketingAnalytics/RawData$ cat ../RawData/customer_data.txt
customer Id , name , city
181 , manav , surat
157 , manan , vadodra
147 , yug , rajkot
manav@LAPTOP-M47MRIJ3:~/MarketingAnalytics/RawData$ cd ..
manav@LAPTOP-M47MRIJ3:~/MarketingAnalytics$ ls -la
total 20
drwxr-xr-x 5 manav manav 4096 Aug  5 12:33 .
drwxr-x--- 6 manav manav 4096 Aug  5 12:32 ..
drwxr-xr-x 2 manav manav 4096 Aug  5 12:33 ProcessedData
drwxr-xr-x 3 manav manav 4096 Aug  5 12:39 RawData
drwxr-xr-x 2 manav manav 4096 Aug  5 12:35 Reports
manav@LAPTOP-M47MRIJ3:~/MarketingAnalytics$ |

```

Key Questions / Analysis

Q1. What is the difference between absolute and relative paths?

Ans:- An absolute path specifies the complete location of a file or directory starting from the root (/) of the filesystem, e.g. /home/analyst01/MarketingAnalytics/RawData. It works the same way regardless of the current working directory. A relative path specifies the location relative to the current working directory, e.g. RawData/sales_data.csv when already inside MarketingAnalytics. Relative paths are shorter and convenient for scripts run from a known location, while absolute paths are safer for scripts that may be executed from anywhere.

Q2. Why are Linux permissions important for file security?

Ans:- Linux uses a permission model (read, write, execute) applied separately to the owner, group, and others for every file and directory. This ensures that sensitive marketing data — such as customer records or sales reports — can only be viewed or modified by authorized users. Permissions prevent accidental or malicious deletion/modification of data, support multi-user collaboration (e.g. giving the analytics team read/write access while restricting outsiders), and form the foundation of the overall security model of the operating system.

Q3. What is the purpose of the grep command?

Ans:- 'grep' (Global Regular Expression Print) searches the content of files for lines matching a specified text pattern or regular expression and prints those matching lines. It is essential for quickly locating specific information — such as a keyword, error message, or data value — inside large log files or reports without opening and reading them manually, e.g. searching all report files for the word 'Conversion' to pull out performance figures instantly.

Q4. How does Linux support efficient file management?

Ans:- Linux provides a rich set of lightweight command-line utilities (mkdir, cp, mv, rm, find, grep, tar, etc.) that can be combined and scripted for automation, supports a hierarchical filesystem for logical organisation of data, offers fine-grained permission and ownership controls, allows redirection and piping to chain commands together, and integrates scheduling tools like cron for recurring tasks such as backups. This combination makes it possible to manage very large volumes of project data quickly, reliably, and in an automatable way.

Q5. Why are backups important in enterprise environments?

Ans:- Backups protect an organisation against data loss caused by hardware failure, accidental deletion, software bugs, or cyberattacks such as ransomware. In an enterprise setting like DataAnalytics Inc., losing raw sales data or client reports could halt business operations, cause financial loss, and damage client trust. Regular, automated backups (e.g. via cron jobs) ensure business continuity, support disaster recovery, and help meet compliance and audit requirements.

Supplementary Problems**1. Create a shell script to automate project directory creation.**

```
manav@LAPTOP-M47MRIJ3:~/MarketingAnalytics/RawData$ cat project_directory.sh
#!/bin/bash

echo "Enter Project Name:"
read project

mkdir -p $project/{RawData,ProcessedData,Reports,Scripts,Backup}

echo "Project Directory Created Successfully."

tree $project
manav@LAPTOP-M47MRIJ3:~/MarketingAnalytics/RawData$
```

output

```
manav@LAPTOP-M47MRIJ3:~/MarketingAnalytics/RawData$ ls
customer.txt  customer_data.txt  project-directory.sh  project_directory.sh
manav@LAPTOP-M47MRIJ3:~/MarketingAnalytics/RawData$ vim project_directory.sh
manav@LAPTOP-M47MRIJ3:~/MarketingAnalytics/RawData$ chmod +x project_directory.sh
manav@LAPTOP-M47MRIJ3:~/MarketingAnalytics/RawData$ ./project_directory.sh
Enter Project Name:
Marketing-Analytics
Project Directory Created Successfully.
./project_directory.sh: line 10: tree: command not found
manav@LAPTOP-M47MRIJ3:~/MarketingAnalytics/RawData$
```

2. Find the top 10 largest files in the system.

```
manav@LAPTOP-M47MRIJ3:~/MarketingAnalytics$ nano largest.sh
manav@LAPTOP-M47MRIJ3:~/MarketingAnalytics$ cat largest.sh
#!/bin/bash

echo "Top 10 Largest Files in the System ---->"

find / -type f -exec du -h {} + 2>/dev/null | sort -rh | head -10
manav@LAPTOP-M47MRIJ3:~/MarketingAnalytics$ ./largest.sh
-bash: ./largest.sh: Permission denied
manav@LAPTOP-M47MRIJ3:~/MarketingAnalytics$ nano largest.sh
manav@LAPTOP-M47MRIJ3:~/MarketingAnalytics$ ./largest.sh
-bash: ./largest.sh: Permission denied
manav@LAPTOP-M47MRIJ3:~/MarketingAnalytics$ cat largest.sh
echo "Top 10 Largest Files in the System ---->"
find / -type f -exec du -h {} + 2>/dev/null | sort -rh | head -10
manav@LAPTOP-M47MRIJ3:~/MarketingAnalytics$
```

3. . Schedule a backup task using cron

```
manav@LAPTOP-M47MRIJ3:~/MarketingAnalytics$ crontab -e
no crontab for manav - using an empty one
Select an editor. To change later, run select-editor again.
 1. /bin/nano          <---- easiest
 2. /usr/bin/vim.basic
 3. /usr/bin/vim.tiny
 4. /bin/ed

Choose 1-4 [1]: 1
No modification made
```

```
manav@LAPTOP-M47MRU3: ~$ nano /tmp/crontab.JTnEcE/crontab
GNU nano 8.7.1 /tmp/crontab.JTnEcE/crontab
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h dom mon dow   command
```

4. Monitor CPU utilization continuously using top/htop.

```
manav@LAPTOP-M47MRU3: ~$ top
top - 13:21:13 up 52 min, 1 user, load average: 0.00, 0.00, 0.00
Tasks: 26 total, 1 running, 24 sleeping, 1 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.6 sy, 0.0 ni, 99.4 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7784.2 total, 7188.4 free, 566.7 used, 180.9 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 7217.5 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM    TIME+  COMMAND
    1 root        20   0   24196   14320  10864 S   0.0   0.2   0:01.11 systemd
    2 root        20   0    3120    1920    1920 S   0.0   0.0   0:00.02 init-systemd(Ub
    7 root        20   0    3120    1792    1792 S   0.0   0.0   0:00.00 init
   47 root        19  -1   50388   15616  14592 S   0.0   0.2   0:00.35 systemd-journal
   84 systemd+    20   0   22400   13952   11520 S   0.0   0.2   0:00.10 systemd-resolve
   90 root        20   0   35252   11776   8832 S   0.0   0.1   0:01.63 systemd-udev
  122 root        20   0    2880    1792    1792 S   0.0   0.0   0:00.08 chronyd-starter
  125 root        20   0    4464    2816    2688 S   0.0   0.0   0:00.01 cron
  131 message+    20   0    8808    4480    4096 S   0.0   0.1   0:00.09 dbus-daemon
  168 root        20   0   44080   27856  13568 S   0.0   0.3   0:00.27 networkd-dispat
  176 root        20   0   18424    8704    7808 S   0.0   0.1   0:00.11 systemd-logind
  201 syslog      20   0   220532  4992    4096 S   0.0   0.1   0:00.11 rsyslogd
  213 root        20   0    5528    2688    2560 S   0.0   0.0   0:00.02agetty
  246 _chrony      20   0   20408    5760    5248 S   0.0   0.1   0:00.12 chronyd
  250 root        20   0    5484    2560    2432 S   0.0   0.0   0:00.00agetty
  251 root        20   0  123444   32128   17664 S   0.0   0.4   0:00.18 unattended-upgr
  255 _chrony      20   0   12080    2396    1792 S   0.0   0.0   0:00.00 chronyd
  337 root        20   0    3136     896     768 S   0.0   0.0   0:00.00 SessionLeader
  338 root        20   0    3136    1160    1024 S   0.0   0.0   0:00.39 Relay(339)
  339 manav      20   0    6364    5504    3840 S   0.0   0.1   0:00.48 bash
  340 root        20   0    8632    5248    4480 S   0.0   0.1   0:00.01 login
  388 manav      20   0   22328   12800   10368 S   0.0   0.2   0:00.32 systemd
  392 manav      20   0   22900    4048    2048 S   0.0   0.1   0:00.00 (sd-pam)
  421 manav      20   0    6128    5376    3840 S   0.0   0.1   0:00.07 bash
 1205 manav      20   0   24000   14208    8320 T   0.0   0.2   0:00.32 vi
 1547 manav      20   0    8164    5376    3328 R   0.0   0.1   0:00.00 top
```

```

manav@LAPTOP-M47MRIJ3: ~
0[ 0.0%] 4[ 0.0%] 8[ 0.0%] 12[ 0.0%]
1[ 0.0%] 5[ 0.0%] 9[ 0.0%] 13[ 0.0%]
2[ 0.0%] 6[ 0.0%] 10[ 0.0%] 14[ 0.0%]
3[ 0.0%] 7[ 0.0%] 11[ 0.0%] 15[ 0.0%]
Mem[|||||] 425M/7.60G Tasks: 28, 14 thr, 0 kthr; 1 running
Swp[ ] 0K/2.00G Load average: 0.07 0.02 0.00
Uptime: 00:54:26

Main I/O
PID USER PRI NI VIRT RES SHR S CPU% MEM% TIME+ Command
1 root 20 0 24196 14320 10864 S 0.0 0.2 0:01.18 /sbin/init
2 root 20 0 3120 1920 1920 S 0.0 0.0 0:00.01 /init
7 root 20 0 3120 1792 1792 S 0.0 0.0 0:00.00 plan9 --control-socket 7 --log-level 4 --server-fd 8
8 root 20 0 3120 1792 0 S 0.0 0.0 0:00.00 plan9 --control-socket 7 --log-level 4 --server-fd 8
9 root 20 0 3120 1920 0 S 0.0 0.0 0:00.00 /init
47 root 19 -1 50388 15616 14592 S 0.0 0.2 0:00.35 /usr/lib/systemd/systemd-journald
84 systemd-re 20 0 22400 13952 11520 S 0.0 0.2 0:00.10 /usr/lib/systemd/systemd-resolved
90 root 20 0 35252 11776 8832 S 0.0 0.1 0:01.64 /usr/lib/systemd/systemd-udevd
122 root 20 0 2880 1792 1792 S 0.0 0.0 0:00.08 /bin/sh /usr/lib/systemd/scripts/chronyd-starter.sh -
125 root 20 0 4464 2816 2688 S 0.0 0.0 0:00.01 /usr/sbin/cron -f -P
131 messagebus 20 0 8808 4480 4096 S 0.0 0.1 0:00.11 @dbus-daemon --system --address=systemd: --nofork --n
168 root 20 0 44080 27856 13568 S 0.0 0.3 0:00.27 /usr/bin/python3 /usr/bin/networkd-dispatcher --run-s
176 root 20 0 18424 8704 7808 S 0.0 0.1 0:00.12 /usr/lib/systemd/systemd-logind
201 syslog 20 0 215M 4992 4096 S 0.0 0.1 0:00.07 /usr/sbin/rsyslogd -n -iNONE
213 root 20 0 5528 2688 2560 S 0.0 0.0 0:00.02 /usr/sbin/agetty --noreset --noclear --issue-file=/et
246 _chrony 20 0 20408 5760 5248 S 0.0 0.1 0:00.13 /usr/sbin/chronyd -n -F 1 -x
247 syslog 20 0 215M 4992 0 S 0.0 0.1 0:00.02 /usr/sbin/rsyslogd -n -iNONE
248 syslog 20 0 215M 4992 0 S 0.0 0.1 0:00.00 /usr/sbin/rsyslogd -n -iNONE
249 syslog 20 0 215M 4992 0 S 0.0 0.1 0:00.01 /usr/sbin/rsyslogd -n -iNONE
250 root 20 0 5484 2560 2432 S 0.0 0.0 0:00.00 /usr/sbin/agetty --noreset --noclear --issue-file=/et
251 root 20 0 120M 32128 17664 S 0.0 0.4 0:00.18 /usr/bin/python3 /usr/share/unattended-upgrades/unatt
255 _chrony 20 0 12080 2396 1792 S 0.0 0.0 0:00.00 /usr/sbin/chronyd -n -F 1 -x
299 root 20 0 120M 32128 0 S 0.0 0.4 0:00.00 /usr/bin/python3 /usr/share/unattended-upgrades/unatt
337 root 20 0 3136 896 768 S 0.0 0.0 0:00.00 /init
338 root 20 0 3136 1160 1024 S 0.0 0.0 0:00.42 /init
339 manav 20 0 6364 5504 3840 S 0.0 0.1 0:00.51 -bash
340 root 20 0 8632 5248 4480 S 0.0 0.1 0:00.01 login -- manav
388 manav 20 0 22328 12800 10368 S 0.0 0.2 0:00.32 /usr/lib/systemd/systemd --user
392 manav 20 0 22900 4048 2048 S 0.0 0.1 0:00.00 (sd-pam)
F1Help F2Setup F3Search F4Filter F5Tree F6SortBy F7Nice F8Nice F9Kill F10Quit

```

5. . Generate a disk usage report for all project directories.

```

manav@LAPTOP-M47MRIJ3:~/MarketingAnalytics$ htop
manav@LAPTOP-M47MRIJ3:~/MarketingAnalytics$ du -sh */ > disk_report.txt
manav@LAPTOP-M47MRIJ3:~/MarketingAnalytics$ cat disk_report.txt
4.0K ProcessedData/
44K RawData/
8.0K Reports/
manav@LAPTOP-M47MRIJ3:~/MarketingAnalytics$

```